2.32 Integration: Standard Functions

Worked Solutions 1–45

1 a)

$\int (e^x + x)dx = e^x + \frac{1}{2}x^2 + c.$

1 b)

$\int e^{2x}dx = \frac{1}{2}e^{2x} + c.$

1 c)

$\int (2e^{3x} - x^2)dx = \frac{2}{3}e^{3x} - \frac{1}{3}x^3 + c.$

1 d)

$\int (\sqrt{e^x} - \frac{9}{x^2})dx = \int (e^{x/2} - 9x^{-2})dx = 2e^{x/2} - \frac{9x^{-1}}{-1} + c = 2e^{x/2} + \frac{9}{x} + c.$

1 e)

$\int \frac{2+e^x}{e^{8x}}dx = \int (\frac{2}{e^{8x}} + \frac{e^x}{e^{8x}})dx = \int (2e^{-8x} + e^{-7x})dx = -\frac{1}{4}e^{-8x} - \frac{1}{7}e^{-7x} + c.$

2 a)

$\frac{dy}{dx} = e^{-3x} \Rightarrow y = -\frac{1}{3}e^{-3x} + c.$

2 b)

$\frac{dy}{dx} = 4 - 2e^{-2x} \Rightarrow y = 4x + e^{-2x} + c.$

2 c)

$\frac{dy}{dx} = \sqrt{x} - \frac{1}{3}e^{x/6} \Rightarrow \frac{dy}{dx} = x^{1/2} - \frac{1}{3}e^{x/6} \Rightarrow y = \frac{2}{3}x^{3/2} - 2e^{x/6} + c.$

2 d)

$\frac{dy}{dx} = x - \frac{6}{e^{4x}} \Rightarrow \frac{dy}{dx} = x - 6e^{-4x} \Rightarrow y = \frac{1}{2}x^2 + \frac{3}{2}e^{-4x} + c.$

3)

$\frac{dy}{dx} = 12e^{6x} \Rightarrow y = 2e^{6x} + c.$ When $x = \ln 2$, $y = 128$: $128 = 2e^{6\ln 2} + c \Rightarrow 128 = 2e^{\ln 2^6} + c \Rightarrow c = 128 - 2e^{\ln 64} \Rightarrow c = 128 - 2 \times 64 = 0.$ $\therefore y = 2e^{6x}.$

4)

$\frac{dy}{dx} = \frac{24}{\sqrt{e^{3x}}} \Rightarrow \frac{dy}{dx} = 24e^{-3x/2} \Rightarrow y = -16e^{-3x/2} + c.$ When $x = \ln 4$, $y = 1$: $1 = -16e^{-\frac{3}{2}\times\ln 4} + c \Rightarrow 1 = -16e^{\ln 4^{-3/2}} + c \Rightarrow 1 = -16e^{\ln(1/8)} + c \Rightarrow 1 = -16 \times \frac{1}{8} + c \Rightarrow c = 3.$ $\therefore y = 3 - 16e^{-3x/2}.$

5 a)

$$\int_0^3 (e^x + 2)dx = [e^x + 2x]_0^3 = (e^3 + 2 \times 3) - (e^0 + 2 \times 0)$$
$$= (e^3 + 6) - 1 = e^3 + 5.$$

5 b)

$$\int_{\ln 2}^{\ln 3} (100 - e^x)dx = [100x - e^x]_{\ln 2}^{\ln 3}$$
$$= (100 \times \ln 3 - e^{\ln 3}) - (100 \times \ln 2 - e^{\ln 2})$$
$$= (100 \ln 3 - 3) - (100 \ln 2 - 2) = 100 \ln 3 - 100 \ln 2 - 1$$
$$= 100(\ln 3 - \ln 2) - 1 = 100 \ln \frac{3}{2} - 1.$$

5 c)

$$\int_{-1}^{1} (24e^{2x/3} - 2x)dx = [36e^{2x/3} - x^2]_{-1}^{1}$$
$$= (36e^{\frac{2}{3} \times 1} - 1^2) - (36e^{\frac{2}{3} \times (-1)} - (-1)^2)$$
$$= (36e^{2/3} - 1) - (36e^{-2/3} - 1) = 36e^{2/3} - 36e^{-2/3}$$
$$= 36(e^{2/3} - e^{-2/3}).$$

5 d)

$$\int_0^{\ln 2} (\frac{5}{e^x} - \frac{2}{e^{2x}})dx = \int_0^{\ln 2} (5e^{-x} - 2e^{-2x})dx = [-5e^{-x} + e^{-2x}]_0^{\ln 2}$$
$$= (-5e^{-\ln 2} + e^{-2 \ln 2}) - (-5e^0 + e^{-2 \times 0})$$
$$= (-5e^{\ln 2^{-1}} + e^{\ln 2^{-2}}) - (-5 + 1) = (-5 \times \tfrac{1}{2} + \tfrac{1}{4}) - (-4) = \frac{7}{4}.$$

6 a)

$$\lim_{\delta x \to 0} \sum_{x=0}^{1} 3e^{18x} \delta x = \int_0^1 3e^{18x} dx = \left[\frac{1}{6} e^{18x}\right]_0^1$$
$$= \frac{1}{6} e^{18 \times 1} - \frac{1}{6} e^{18 \times 0} = \frac{1}{6} e^{18} - \frac{1}{6} e^0$$
$$= \frac{1}{6} e^{18} - \frac{1}{6} = \frac{1}{6}(e^{18} - 1).$$

6 b)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=2}^{\ln 8}\frac{1}{\sqrt[3]{e^x}}\delta x &= \int_2^{\ln 8} e^{-x/3}dx = [-3e^{-x/3}]_2^{\ln 8}\\ &= (-3e^{-\frac{1}{3}\times\ln 8}) - (-3e^{-\frac{1}{3}\times 2})\\ &= (-3e^{\ln 8^{-1/3}}) - (-3e^{-2/3}) = (-3\times\tfrac{1}{2}) - (-3e^{-2/3})\\ &= -\frac{3}{2} + 3e^{-2/3}.\end{aligned}$$

7)

$$\begin{aligned}\int_{\ln(1/3)}^{\ln 3}\sqrt{e^x}dx &= \int_{\ln(1/3)}^{\ln 3} e^{x/2}dx = [2e^{x/2}]_{\ln(1/3)}^{\ln 3}\\ &= 2e^{\frac{1}{2}\ln 3} - 2e^{\frac{1}{2}\ln(1/3)} = 2e^{\ln 3^{1/2}} - 2e^{\ln(1/3)^{1/2}}\\ &= 2\sqrt{3} - 2\times\frac{1}{\sqrt{3}} = 2\sqrt{3} - \frac{2}{\sqrt{3}}\\ &= 2\sqrt{3} - \frac{2\sqrt{3}}{3} = \frac{4\sqrt{3}}{3}.\end{aligned}$$

8)

$2-e^x = e^{2x}-4 \Rightarrow e^{2x}+e^x-6=0 \Rightarrow (e^x+3)(e^x-2)=0 \Rightarrow e^x = 2$ as $e^x \neq -3$. $e^x = 2 \Rightarrow x = \ln 2$.

$$\begin{aligned}\int_0^{\ln 2}(2-e^x)dx - \int_0^{\ln 2}(e^{2x}-4)dx &= \int_0^{\ln 2}(6-e^x-e^{2x})dx = [6x - e^x - \tfrac{1}{2}e^{2x}]_0^{\ln 2}\\ &= (6\times\ln 2 - e^{\ln 2} - \tfrac{1}{2}e^{2\ln 2}) - (6\times 0 - e^0 - \tfrac{1}{2}e^{2\times 0})\\ &= (6\ln 2 - 2 - \tfrac{1}{2}e^{\ln 2^2}) - (0 - 1 - \tfrac{1}{2}e^0)\\ &= (6\ln 2 - 2 - \tfrac{1}{2}\times 4) - (-1 - \tfrac{1}{2})\\ &= (6\ln 2 - 4) - (-\tfrac{3}{2}) = 6\ln 2 - \frac{5}{2}.\end{aligned}$$

9)

$y = e^2x - e^x \Rightarrow \dfrac{dy}{dx} = e^2 - e^x$. Turning points exist when $\frac{dy}{dx} = 0$: $e^2 - e^x = 0 \Rightarrow e^x = e^2 \Rightarrow x = 2$.
$\therefore B$, C and D have coordinates $(1,0)$, $(2,0)$ and $(3,0)$ respectively.

$$\begin{aligned}\int_1^3 (e^2x - e^x)dx &= [\tfrac{1}{2}e^2x^2 - e^x]_1^3\\ &= (\tfrac{1}{2}e^2\times 3^2 - e^3) - (\tfrac{1}{2}e^2\times 1^2 - e^1)\\ &= (\tfrac{9}{2}e^2 - e^3) - (\tfrac{1}{2}e^2 - e) = 4e^2 - e^3 + e.\end{aligned}$$

10 a)

$\int(\sin x + \cos x)dx = -\cos x + \sin x + c$.

10 b)

$\int(2\sin x - \cos x)dx = -2\cos x - \sin x + c.$

10 c)

$\int(2\sin 2x + \cos 4x)dx = -\cos 2x + \frac{1}{4}\sin 4x + c.$

10 d)

$\int(3\sin \frac{1}{2}x - 6\cos \frac{2}{3}x)dx = -6\cos \frac{1}{2}x - 9\sin \frac{2}{3}x + c.$

11 a)

$\frac{dy}{dx} = 3\cos 6x \Rightarrow y = \frac{1}{2}\sin 6x + c.$

11 b)

$\frac{dy}{dx} = \frac{3}{x^2} - 4\sin \frac{1}{4}x \Rightarrow \frac{dy}{dx} = 3x^{-2} - 4\sin \frac{1}{4}x \Rightarrow y = \frac{3x^{-1}}{-1} + 16\cos \frac{1}{4}x + c \Rightarrow y = -\frac{3}{x} + 16\cos \frac{1}{4}x + c.$

11 c)

$\frac{dy}{dx} = \sqrt[3]{x} - \frac{2}{3}\cos \frac{5}{6}x \Rightarrow \frac{dy}{dx} = x^{1/3} - \frac{2}{3}\cos \frac{5}{6}x \Rightarrow y = \frac{x^{4/3}}{4/3} - \frac{4}{5}\sin \frac{5}{6}x + c \Rightarrow y = \frac{3}{4}x^{4/3} - \frac{4}{5}\sin \frac{5}{6}x + c.$

12)

$\frac{dy}{dx} = -4\sin 3x \Rightarrow y = \frac{4}{3}\cos 3x + c.$ When $x = \frac{\pi}{6}$, $y = 3$: $3 = \frac{4}{3}\cos(3 \times \frac{\pi}{6}) + c \Rightarrow 3 = \frac{4}{3}\cos \frac{\pi}{2} + c \Rightarrow 3 = 0 + c \Rightarrow c = 3.$ $\therefore y = \frac{4}{3}\cos 3x + 3.$

13)

$\frac{dy}{dx} = 2\sin \frac{1}{3}x - 3\cos \frac{1}{3}x \Rightarrow y = -6\cos \frac{1}{3}x - 9\sin \frac{1}{3}x + c.$ When $x = \pi$, $y = -\frac{9\sqrt{3}}{2} - 3$: $-\frac{9\sqrt{3}}{2} - 3 = -6\cos(\frac{1}{3} \times \pi) - 9\sin(\frac{1}{3} \times \pi) + c \Rightarrow -\frac{9\sqrt{3}}{2} - 3 = -6\cos \frac{\pi}{3} - 9\sin \frac{\pi}{3} + c \Rightarrow -\frac{9\sqrt{3}}{2} - 3 = -3 - \frac{9\sqrt{3}}{2} + c \Rightarrow c = 0.$ $\therefore y = -6\cos \frac{1}{3}x - 9\sin \frac{1}{3}x.$

14 a)

$\int_0^{\pi/2} \sin x\, dx = [-\cos x]_0^{\pi/2} = (-\cos \frac{\pi}{2}) - (-\cos 0) = 0 - (-1) = 1.$

14 b)

$$\int_0^{\pi/4} \cos 2x\, dx = [\tfrac{1}{2}\sin 2x]_0^{\pi/4} = \frac{1}{2}\sin(2 \times \tfrac{\pi}{4}) - \frac{1}{2}\sin(2 \times 0)$$
$$= \frac{1}{2}\sin \frac{\pi}{2} - \frac{1}{2}\sin 0 = \frac{1}{2} - 0 = \frac{1}{2}.$$

14 c)

$$\begin{aligned}\int_{-5\pi/6}^{5\pi/3} 3\sin\tfrac{1}{5}x\,dx &= [-15\cos\tfrac{1}{5}x]_{-5\pi/6}^{5\pi/3}\\ &= -15\cos(\tfrac{1}{5}\times\tfrac{5\pi}{3}) - (-15\cos(\tfrac{1}{5}\times-\tfrac{5\pi}{6}))\\ &= -15\cos\frac{\pi}{3} - (-15\cos(-\tfrac{\pi}{6})) = -\frac{15}{2} - (-\tfrac{15\sqrt{3}}{2})\\ &= \frac{15\sqrt{3}}{2} - \frac{15}{2} = \frac{15}{2}(\sqrt{3}-1).\end{aligned}$$

14 d)

$$\begin{aligned}\int_{\pi/8}^{\pi/4}(5\sin 4x - 6\cos 4x)dx &= [-\tfrac{5}{4}\cos 4x - \tfrac{3}{2}\sin 4x]_{\pi/8}^{\pi/4}\\ &= (-\tfrac{5}{4}\cos(4\times\tfrac{\pi}{4}) - \tfrac{3}{2}\sin(4\times\tfrac{\pi}{4}))\\ &\quad - (-\tfrac{5}{4}\cos(4\times\tfrac{\pi}{8}) - \tfrac{3}{2}\sin(4\times\tfrac{\pi}{8}))\\ &= (-\tfrac{5}{4}\cos\pi - \tfrac{3}{2}\sin\pi) - (-\tfrac{5}{4}\cos\tfrac{\pi}{2} - \tfrac{3}{2}\sin\tfrac{\pi}{2})\\ &= (\tfrac{5}{4}-0) - (0-\tfrac{3}{2}) = \frac{11}{4}.\end{aligned}$$

15 a)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=0}^{\pi/12}(8\sin 4x - 16\cos 4x)\delta x &= \int_0^{\pi/12}(8\sin 4x - 16\cos 4x)dx = [-2\cos 4x - 4\sin 4x]_0^{\pi/12}\\ &= (-2\cos(4\times\tfrac{\pi}{12}) - 4\sin(4\times\tfrac{\pi}{12}))\\ &\quad - (-2\cos(4\times 0) - 4\sin(4\times 0))\\ &= (-2\cos\tfrac{\pi}{3} - 4\sin\tfrac{\pi}{3}) - (-2\cos 0 - 4\sin 0)\\ &= (-1-2\sqrt{3}) - (-2-0) = 1-2\sqrt{3}.\end{aligned}$$

15 b)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=9\pi}^{10\pi}(\tfrac{1}{8}\sin\tfrac{1}{4}x - \tfrac{1}{16}\cos\tfrac{1}{4}x)\delta x &= \int_{9\pi}^{10\pi}(\tfrac{1}{8}\sin\tfrac{1}{4}x - \tfrac{1}{16}\cos\tfrac{1}{4}x)dx\\ &= [-\tfrac{1}{2}\cos\tfrac{1}{4}x - \tfrac{1}{4}\sin\tfrac{1}{4}x]_{9\pi}^{10\pi}\\ &= (-\tfrac{1}{2}\cos(\tfrac{1}{4}\times 10\pi) - \tfrac{1}{4}\sin(\tfrac{1}{4}\times 10\pi))\\ &\quad - (-\tfrac{1}{2}\cos(\tfrac{1}{4}\times 9\pi) - \tfrac{1}{4}\sin(\tfrac{1}{4}\times 9\pi))\\ &= (-\tfrac{1}{2}\cos\tfrac{5\pi}{2} - \tfrac{1}{4}\sin\tfrac{5\pi}{2}) - (-\tfrac{1}{2}\cos\tfrac{9\pi}{4} - \tfrac{1}{4}\sin\tfrac{9\pi}{4})\\ &= (0-\tfrac{1}{4}) - (-\tfrac{\sqrt{2}}{4} - \tfrac{\sqrt{2}}{8}) = -\frac{1}{4} - (-\tfrac{3\sqrt{2}}{8}) = \frac{3\sqrt{2}}{8} - \frac{1}{4}.\end{aligned}$$

16)

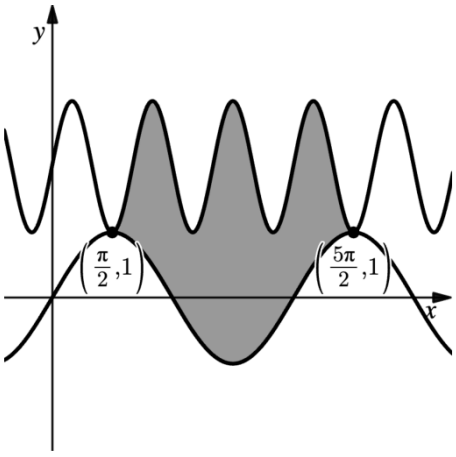
$$\begin{aligned}\int_{\pi}^{4\pi}(3+2\sin\tfrac{1}{2}x)dx &= [3x-4\cos\tfrac{1}{2}x]_{\pi}^{4\pi}\\ &= (3\times 4\pi-4\cos(\tfrac{1}{2}\times 4\pi))-(3\times\pi-4\cos(\tfrac{1}{2}\times\pi))\\ &= (12\pi-4\cos 2\pi)-(3\pi-4\cos\tfrac{\pi}{2})\\ &= (12\pi-4)-(3\pi-0)=9\pi-4.\end{aligned}$$

17)

$$\begin{aligned}\int_{\pi/2}^{9\pi/2}(\cos\tfrac{1}{3}x+2)dx-\int_{\pi/2}^{9\pi/2}(\sin\tfrac{1}{3}x-2)dx &= \int_{\pi/2}^{9\pi/2}(\cos\tfrac{1}{3}x-\sin\tfrac{1}{3}x+4)dx\\ &= [3\sin\tfrac{1}{3}x+3\cos\tfrac{1}{3}x+4x]_{\pi/2}^{9\pi/2}\\ &= (3\sin(\tfrac{1}{3}\times\tfrac{9\pi}{2})+3\cos(\tfrac{1}{3}\times\tfrac{9\pi}{2})+4\times\tfrac{9\pi}{2})\\ &\quad-(3\sin(\tfrac{1}{3}\times\tfrac{\pi}{2})+3\cos(\tfrac{1}{3}\times\tfrac{\pi}{2})+4\times\tfrac{\pi}{2})\\ &= (3\sin\tfrac{3\pi}{2}+3\cos\tfrac{3\pi}{2}+18\pi)-(3\sin\tfrac{\pi}{6}+3\cos\tfrac{\pi}{6}+2\pi)\\ &= (-3+0+18\pi)-(\tfrac{3}{2}+\tfrac{3\sqrt{3}}{2}+2\pi)=16\pi-\frac{3\sqrt{3}}{2}-\frac{9}{2}.\end{aligned}$$

18)

The top curve has a period of $\frac{2\pi}{3}$, so the curve has equation $y=2+\sin 3x$.



$$\begin{aligned}\int_{\pi/2}^{5\pi/2}(2+\sin 3x)dx-\int_{\pi/2}^{5\pi/2}\sin x\,dx &= \int_{\pi/2}^{5\pi/2}(2+\sin 3x-\sin x)dx\\ &= [2x-\tfrac{1}{3}\cos 3x+\cos x]_{\pi/2}^{5\pi/2}\\ &= (2\times\tfrac{5\pi}{2}-\tfrac{1}{3}\cos(3\times\tfrac{5\pi}{2})+\cos\tfrac{5\pi}{2})\\ &\quad-(2\times\tfrac{\pi}{2}-\tfrac{1}{3}\cos(3\times\tfrac{\pi}{2})+\cos\tfrac{\pi}{2})\\ &= (5\pi-\tfrac{1}{3}\cos\tfrac{15\pi}{2}+\cos\tfrac{5\pi}{2})-(\pi-\tfrac{1}{3}\cos\tfrac{3\pi}{2}+\cos\tfrac{\pi}{2})\\ &= (5\pi-0+0)-(\pi-0+0)=4\pi.\end{aligned}$$

19 a)

$\int 3^x dx = \frac{1}{\ln 3}3^x + c.$

19 b)

$\int (3^{2x} + 4^{3x})dx = \frac{1}{2\ln 3}3^{2x} + \frac{1}{3\ln 4}4^{3x} + c = \frac{1}{\ln 9}9^x + \frac{1}{\ln 64}64^x + c.$

19 c)

$\int (\frac{1}{5^x} - \frac{2}{2^x})dx = \int (5^{-x} - 2 \times 2^{-x})dx = -\frac{1}{\ln 5}5^{-x} - 2 \times -\frac{1}{\ln 2}2^{-x} + c = -\frac{1}{(\ln 5)5^x} + \frac{2}{(\ln 2)2^x} + c.$

19 d)

$\int \frac{3^x - 4}{9^x}dx = \int (\frac{3^x}{9^x} - \frac{4}{9^x})dx = \int (\frac{3^x}{3^{2x}} - \frac{4}{9^x})dx = \int (3^{-x} - 4 \times 9^{-x})dx = -\frac{1}{\ln 3}3^{-x} - 4 \times -\frac{1}{\ln 9}9^{-x} + c = -\frac{1}{(\ln 3)3^x} + \frac{4}{(\ln 9)9^x} + c.$

20 a)

$\frac{dy}{dx} = 8^x \Rightarrow y = \frac{1}{\ln 8}8^x + c.$

20 b)

$\frac{dy}{dx} = x - 3^x \ln 3 \Rightarrow y = \frac{1}{2}x^2 - \frac{1}{\ln 3}3^x \times \ln 3 + c \Rightarrow y = \frac{1}{2}x^2 - 3^x + c.$

20 c)

$\frac{dy}{dx} = 6^{3x} - \frac{1}{6^{3x}} \Rightarrow \frac{dy}{dx} = 6^{3x} - 6^{-3x} \Rightarrow y = \frac{1}{3\ln 6}6^{3x} - (-\frac{1}{3\ln 6}6^{-3x}) \Rightarrow y = \frac{1}{3\ln 6}6^{3x} + \frac{1}{3\ln 6}6^{-3x}.$

20 d)

$\frac{dy}{dx} = \frac{16^x - 4}{2^x} \Rightarrow \frac{dy}{dx} = \frac{16^x}{2^x} - \frac{4}{2^x} \Rightarrow \frac{dy}{dx} = \frac{2^{4x}}{2^x} - \frac{4}{2^x} \Rightarrow \frac{dy}{dx} = 2^{3x} - 4 \times 2^{-x} \Rightarrow y = \frac{1}{3\ln 2}2^{3x} - 4 \times -\frac{1}{\ln 2}2^{-x} \Rightarrow y = \frac{1}{3\ln 2}2^{3x} + \frac{4}{\ln 2}2^{-x}.$

21)

$\frac{dy}{dx} = 25^x \Rightarrow y = \frac{1}{\ln 25}25^x.$ When $x = 0$, $y = 20$: $20 = \frac{1}{\ln 25} \times 25^0 + c \Rightarrow c = 20 - \frac{1}{\ln 25}.$
$\therefore y = \frac{1}{\ln 25}25^x + 20 - \frac{1}{\ln 25} \Rightarrow y = \frac{25^x + 20\ln 25 - 1}{\ln 25}.$

22)

$\frac{dy}{dx} = \frac{1}{4^x} \Rightarrow \frac{dy}{dx} = 4^{-x} \Rightarrow y = -\frac{1}{\ln 4}4^{-x} + c.$ When $x = 0$, $y = \frac{1}{\ln 4}$: $\frac{1}{\ln 4} = -\frac{1}{\ln 4}4^0 + c \Rightarrow c = \frac{2}{\ln 4}.$

$$y = -\frac{1}{\ln 4}4^{-x} + \frac{2}{\ln 4} = -\frac{1}{2\ln 2}2^{-2x} + \frac{2}{2\ln 2}$$
$$= \frac{2 - 2^{-2x}}{2\ln 2} = \frac{2(1 - 2^{-2x-1})}{2\ln 2} = \frac{1 - 2^{-2x-1}}{\ln 2}.$$

23 a)

$$\int_{-1}^{1} 3^x dx = [\tfrac{1}{\ln 3}3^x]_{-1}^{1} = (\tfrac{1}{\ln 3} \times 3^1) - (\tfrac{1}{\ln 3} \times 3^{-1})$$
$$= \frac{3}{\ln 3} - \frac{1}{3\ln 3} = \frac{8}{3\ln 3}.$$

23 b)

$$\begin{aligned}\int_0^2 (2^x - 4^x)dx &= [\tfrac{1}{\ln 2}2^x - \tfrac{1}{\ln 4}4^x]_0^2 \\ &= (\tfrac{1}{\ln 2} \times 2^2 - \tfrac{1}{\ln 4} \times 4^2) - (\tfrac{1}{\ln 2} \times 2^0 - \tfrac{1}{\ln 4} \times 4^0) \\ &= (\tfrac{4}{\ln 2} - \tfrac{16}{\ln 4}) - (\tfrac{1}{\ln 2} - \tfrac{1}{\ln 4}) \\ &= (\tfrac{4}{\ln 2} - \tfrac{16}{2\ln 2}) - (\tfrac{1}{\ln 2} - \tfrac{1}{2\ln 2}) \\ &= (\tfrac{4}{\ln 2} - \tfrac{8}{\ln 2}) - (\tfrac{1}{\ln 2} - \tfrac{1}{2\ln 2}) \\ &= -\frac{4}{\ln 2} - \frac{1}{2\ln 2} = -\frac{9}{2\ln 2}.\end{aligned}$$

23 c)

$$\begin{aligned}\int_{-1/3}^{1/3} (5^{3x} - x)dx &= [\tfrac{1}{3\ln 5}5^{3x} - \tfrac{1}{2}x^2]_{-1/3}^{1/3} \\ &= (\tfrac{1}{3\ln 5} \times 5^{3\times 1/3} - \tfrac{1}{2} \times (\tfrac{1}{3})^2) \\ &\quad - (\tfrac{1}{3\ln 5} \times 5^{3\times -1/3} - \tfrac{1}{2} \times (-\tfrac{1}{3})^2) \\ &= (\tfrac{5}{3\ln 5} - \tfrac{1}{18}) - (\tfrac{1}{15\ln 5} - \tfrac{1}{18}) = \frac{5}{3\ln 5} - \frac{1}{15\ln 5} = \frac{8}{5\ln 5}.\end{aligned}$$

23 d)

$$\begin{aligned}\int_1^2 \frac{2 - 4^x}{4^x}dx &= \int_1^2 (\tfrac{2}{4^x} - \tfrac{4^x}{4^x})dx = \int_1^2 (2 \times 4^{-x} - 1)dx \\ &= [2 \times -\tfrac{1}{\ln 4}4^{-x} - x]_1^2 = [-\tfrac{2}{\ln 4}4^{-x} - x]_1^2 \\ &= (-\tfrac{2}{\ln 4} \times 4^{-2} - 2) - (-\tfrac{2}{\ln 4} \times 4^{-1} - 1) \\ &= (-\tfrac{1}{8\ln 4} - 2) - (-\tfrac{1}{2\ln 4} - 1) = \frac{3}{8\ln 4} - 1.\end{aligned}$$

24 a)

$$\begin{aligned}\lim_{\delta x \to 0} \sum_{x=5}^{10} 5^{0.2x}\delta x &= \int_5^{10} 5^{0.2x}dx = \int_5^{10} 5^{x/5}dx = [\tfrac{5}{\ln 5}5^{x/5}]_5^{10} \\ &= (\tfrac{5}{\ln 5} \times 5^{\frac{1}{5}\times 10}) - (\tfrac{5}{\ln 5} \times 5^{\frac{1}{5}\times 5}) \\ &= (\tfrac{5}{\ln 5} \times 25) - (\tfrac{5}{\ln 5} \times 5) = \frac{125}{\ln 5} - \frac{25}{\ln 5} = \frac{100}{\ln 5}.\end{aligned}$$

24 b)

$$\lim_{\delta x\to 0}\sum_{x=1/10}^{1/5}\frac{10}{2^{5x}}\delta x=\int_{1/10}^{1/5}10\times 2^{-5x}dx$$
$$=[10\times -\tfrac{1}{5\ln 2}2^{-5x}]_{1/10}^{1/5}=[-\tfrac{2}{\ln 2}2^{-5x}]_{1/10}^{1/5}$$
$$=(-\tfrac{2}{\ln 2}\times 2^{-5\times 1/5})-(-\tfrac{2}{\ln 2}\times 2^{-5\times 1/10})$$
$$=(-\tfrac{2}{\ln 2}\times \tfrac{1}{2})-(-\tfrac{2}{\ln 2}\times \tfrac{1}{\sqrt{2}})$$
$$=(-\tfrac{1}{\ln 2})-(-\tfrac{2}{\ln 2}\times \tfrac{\sqrt{2}}{2})=-\frac{1}{\ln 2}-(-\tfrac{\sqrt{2}}{\ln 2})$$
$$=-\frac{1}{\ln 2}+\frac{\sqrt{2}}{\ln 2}=\frac{\sqrt{2}-1}{\ln 2}.$$

25)

$$\int_{1/2}^{1}(4x-4^x)dx=[2x^2-\tfrac{1}{\ln 4}4^x]_{1/2}^{1}$$
$$=(2\times 1^2-\tfrac{1}{\ln 4}\times 4^1)-(2\times(\tfrac{1}{2})^2-\tfrac{1}{\ln 4}\times 4^{1/2})$$
$$=(2-\tfrac{4}{\ln 4})-(\tfrac{1}{2}-\tfrac{2}{\ln 4})=\frac{3}{2}-\frac{2}{\ln 4}$$
$$=\frac{3}{2}-\frac{2}{2\ln 2}=\frac{3}{2}-\frac{1}{\ln 2}.$$

26)

$2^x+5=4^x-1\Rightarrow 2^x+5=2^{2x}-1\Rightarrow 2^{2x}-2^x-6=0\Rightarrow(2^x-3)(2^x+2)=0\Rightarrow 2^x=3$ as $2^x\neq -2$.
$2^x=3\Rightarrow x=\log_2 3$.

$$\int_0^{\log_2 3}(2^x+5)dx-\int_0^{\log_2 3}(4^x-1)dx=\int_0^{\log_2 3}(2^x-4^x+6)dx=[\tfrac{1}{\ln 2}2^x-\tfrac{1}{\ln 4}4^x+6x]_0^{\log_2 3}$$
$$=(\tfrac{1}{\ln 2}2^{\log_2 3}-\tfrac{1}{\ln 4}4^{\log_2 3}+6\log_2 3)$$
$$-(\tfrac{1}{\ln 2}2^0-\tfrac{1}{\ln 4}4^0+6\times 0)$$
$$=(\tfrac{3}{\ln 2}-\tfrac{9}{\ln 4}+6\log_2 3)-(\tfrac{1}{\ln 2}-\tfrac{1}{\ln 4})$$
$$=\frac{2}{\ln 2}-\frac{8}{\ln 4}+6\log_2 3.$$

27)

$y=5^x\Rightarrow\dfrac{dy}{dx}=(\ln 5)5^x$. $\left.\dfrac{dy}{dx}\right|_{x=1}=(\ln 5)\times 5^1=5\ln 5$. Tangent: $y-5=5\ln 5(x-1)$. When $y=0$:
$-5=5\ln 5(x-1)\Rightarrow x-1=-\dfrac{1}{\ln 5}\Rightarrow x=1-\dfrac{1}{\ln 5}$.

$$\text{Area}=\int_{1-1/\ln 5}^{1}5^x dx-\frac{1}{2}\times\left(1-(1-\tfrac{1}{\ln 5})\right)\times 5$$
$$=[\tfrac{1}{\ln 5}5^x]_{1-1/\ln 5}^{1}-\frac{5}{2\ln 5}$$
$$=(\tfrac{1}{\ln 5}\times 5^1)-(\tfrac{1}{\ln 5}\times 5^{1-1/\ln 5})-\frac{5}{2\ln 5}$$

$$= \frac{5}{\ln 5} - (\tfrac{1}{\ln 5} \times 5^1 \times 5^{-1/\ln 5}) - \frac{5}{2\ln 5}$$
$$= \frac{5}{\ln 5} - (\tfrac{5}{\ln 5} \times (e^{\ln 5})^{-1/\ln 5}) - \frac{5}{2\ln 5}$$
$$= \frac{5}{\ln 5} - (\tfrac{5}{\ln 5} \times e^{-1}) - \frac{5}{2\ln 5}$$
$$= \frac{5}{\ln 5} - \frac{5}{e\ln 5} - \frac{5}{2\ln 5} = \frac{5e-5}{e\ln 5} - \frac{5}{2\ln 5}$$
$$= \frac{10e-10}{2e\ln 5} - \frac{5e}{2e\ln 5} = \frac{5e-10}{2e\ln 5}.$$

28 a)

$\int \frac{1}{x}dx = \ln|x| + c.$

28 b)

$\int(\frac{3}{x} - \frac{x}{3})dx = 3\ln|x| - \frac{1}{6}x^2 + c.$

28 c)

$\int(\frac{5}{3x} + \frac{2}{5x^2})dx = \int(\frac{5}{3} \times \frac{1}{x} + \frac{2}{5}x^{-2})dx = \frac{5}{3}\ln|x| + \frac{2}{5} \times \frac{x^{-1}}{-1} + c = \frac{5}{3}\ln|x| - \frac{2}{5x} + c.$

Alternatively (by reversing the chain rule): $\int(\frac{5}{3x} + \frac{2}{5x^2})dx = \frac{5}{3}\ln|3x| + \frac{2}{5} \times \frac{x^{-1}}{-1} + c = \frac{5}{3}\ln|3x| - \frac{2}{5x} + c.$

28 d)

$\int \frac{3+8x}{4x}dx = \int(\frac{3}{4x} + \frac{8x}{4x})dx = \int(\frac{3}{4} \times \frac{1}{x} + 2)dx = \frac{3}{4}\ln|x| + 2x + c.$

Alternatively (by reversing the chain rule): $\int \frac{3+8x}{4x}dx = \int(\frac{3}{4x} + \frac{8x}{4x})dx = \int(\frac{3}{4x} + 2)dx = \frac{3}{4}\ln|4x| + 2x + c.$

29 a)

$\frac{dy}{dx} = \frac{4}{x} - \frac{4}{x^2} \Rightarrow \frac{dy}{dx} = \frac{4}{x} - 4x^{-2} \Rightarrow y = 4\ln|x| - \frac{4x^{-1}}{-1} + c \Rightarrow y = 4\ln|x| + \frac{4}{x} + c.$

29 b)

$\frac{dy}{dx} = 2 - \frac{7}{4x} \Rightarrow \frac{dy}{dx} = 2 - \frac{7}{4} \times \frac{1}{x} \Rightarrow y = 2x - \frac{7}{4}\ln|x| + c.$

Alternatively (by reversing the chain rule): $\frac{dy}{dx} = 2 - \frac{7}{4x} \Rightarrow y = 2x - \frac{7}{4}\ln|4x| + c.$

29 c)

$\frac{dy}{dx} = \frac{3\sqrt{x}+1}{x^{3/2}} \Rightarrow \frac{dy}{dx} = \frac{3x^{1/2}}{x^{3/2}} + \frac{1}{x^{3/2}} \Rightarrow \frac{dy}{dx} = \frac{3}{x} + x^{-3/2} \Rightarrow y = 3\ln|x| + \frac{x^{-1/2}}{-1/2} + c \Rightarrow y = 3\ln|x| - 2x^{-1/2} + c \Rightarrow y = 3\ln|x| - \frac{2}{\sqrt{x}} + c.$

29 d)

$\frac{dy}{dx} = \frac{8x-3}{5x^2} \Rightarrow \frac{dy}{dx} = \frac{8x}{5x^2} - \frac{3}{5x^2} \Rightarrow \frac{dy}{dx} = \frac{8}{5} \times \frac{1}{x} - \frac{3}{5}x^{-2} \Rightarrow y = \frac{8}{5}\ln|x| - \frac{\frac{3}{5}x^{-1}}{-1} + c \Rightarrow y = \frac{8}{5}\ln|x| + \frac{3}{5}x^{-1} + c \Rightarrow y = \frac{8}{5}\ln|x| + \frac{3}{5x} + c.$

30)

$\frac{dy}{dx} = \frac{5}{x} \Rightarrow y = 5\ln|x| + c$. When $x = e^5$, $y = 1$: $1 = 5\ln e^5 + c \Rightarrow 1 = 5 \times 5 + c \Rightarrow c = -24$.
$\therefore y = 5\ln|x| - 24 \Rightarrow y = -24 + 5\ln|x|$.

31)

$\frac{dy}{dx} = \frac{3-10x}{x} \Rightarrow \frac{dy}{dx} = \frac{3}{x} - \frac{10x}{x} \Rightarrow \frac{dy}{dx} = \frac{3}{x} - 10 \Rightarrow y = 3\ln|x| - 10x + c$. When $x = 1$, $y = 1$:
$1 = 3\ln 1 - 10\times 1 + c \Rightarrow 1 = 3\times 0 - 10 + c \Rightarrow c = 11$. $\therefore y = 3\ln|x| - 10x + 11 \Rightarrow y = -10x + 11 + 3\ln|x|$.

32 a)

$\int_1^2 \frac{9}{x}dx = [9\ln|x|]_1^2 = 9\ln|2| - 9\ln|1| = 9\ln 2 - 0 = 9\ln 2$.

32 b)

$$\begin{aligned}\int_{-3}^{-1} \frac{8}{x}dx &= [8\ln|x|]_{-3}^{-1} = 8\ln|-1| - 8\ln|-3| \\ &= 8\ln 1 - 8\ln 3 = 0 - 8\ln 3 = -8\ln 3 = 8\ln 3^{-1} = 8\ln\frac{1}{3}.\end{aligned}$$

32 c)

$$\begin{aligned}\int_1^4 \frac{2+x}{10x}dx &= \int_1^4 (\tfrac{2}{10x} + \tfrac{x}{10x})dx = \int_1^4 (\tfrac{1}{5}\times\tfrac{1}{x} + \tfrac{1}{10})dx \\ &= [\tfrac{1}{5}\ln|x| + \tfrac{1}{10}x]_1^4 \\ &= (\tfrac{1}{5}\ln|4| + \tfrac{1}{10}\times 4) - (\tfrac{1}{5}\ln|1| + \tfrac{1}{10}\times 1) \\ &= (\tfrac{1}{5}\ln 4 + \tfrac{2}{5}) - (\tfrac{1}{5}\ln 1 + \tfrac{1}{10}) = (\tfrac{1}{5}\ln 4 + \tfrac{2}{5}) - (0 + \tfrac{1}{10}) \\ &= \frac{1}{5}\ln 4 + \frac{3}{10}.\end{aligned}$$

Alternatively (by reversing the chain rule):

$$\begin{aligned}\int_1^4 \frac{2+x}{10x}dx &= \int_1^4 (\tfrac{2}{10x} + \tfrac{x}{10x})dx = \int_1^4 (\tfrac{1}{5x} + \tfrac{1}{10})dx \\ &= [\tfrac{1}{5}\ln|5x| + \tfrac{1}{10}x]_1^4 \\ &= (\tfrac{1}{5}\ln|5\times 4| + \tfrac{1}{10}\times 4) - (\tfrac{1}{5}\ln|5\times 1| + \tfrac{1}{10}\times 1) \\ &= (\tfrac{1}{5}\ln 20 + \tfrac{2}{5}) - (\tfrac{1}{5}\ln 5 + \tfrac{1}{10}) \\ &= \frac{1}{5}(\ln 20 - \ln 5) + \frac{3}{10} = \frac{1}{5}\ln\frac{20}{5} + \frac{3}{10} = \frac{1}{5}\ln 4 + \frac{3}{10}.\end{aligned}$$

32 d)

$$\begin{aligned}\int_{-3}^{-2} \frac{3-x}{5x}dx &= \int_{-3}^{-2} (\tfrac{3}{5x} - \tfrac{x}{5x})dx = \int_{-3}^{-2} (\tfrac{3}{5}\times\tfrac{1}{x} - \tfrac{1}{5})dx \\ &= [\tfrac{3}{5}\ln|x| - \tfrac{1}{5}x]_{-3}^{-2} \\ &= (\tfrac{3}{5}\ln|-2| - \tfrac{1}{5}\times -2) - (\tfrac{3}{5}\ln|-3| - \tfrac{1}{5}\times -3)\end{aligned}$$

$$= (\tfrac{3}{5}\ln 2 + \tfrac{2}{5}) - (\tfrac{3}{5}\ln 3 + \tfrac{3}{5}) = \frac{3}{5}(\ln 2 - \ln 3) - \frac{1}{5}$$
$$= \frac{3}{5}\ln\frac{2}{3} - \frac{1}{5}.$$

Alternatively (by reversing the chain rule):

$$\int_{-3}^{-2} \frac{3-x}{5x}dx = \int_{-3}^{-2} (\tfrac{3}{5x} - \tfrac{x}{5x})dx = \int_{-3}^{-2} (\tfrac{3}{5x} - \tfrac{1}{5})dx$$
$$= [\tfrac{3}{5}\ln|5x| - \tfrac{1}{5}x]_{-3}^{-2}$$
$$= (\tfrac{3}{5}\ln|5\times -2| - \tfrac{1}{5}\times -2) - (\tfrac{3}{5}\ln|5\times -3| - \tfrac{1}{5}\times -3)$$
$$= (\tfrac{3}{5}\ln 10 + \tfrac{2}{5}) - (\tfrac{3}{5}\ln 15 + \tfrac{3}{5}) = \frac{3}{5}(\ln 10 - \ln 15) - \frac{1}{5}$$
$$= \frac{3}{5}\ln\frac{10}{15} - \frac{1}{5} = \frac{3}{5}\ln\frac{2}{3} - \frac{1}{5}.$$

33 a)

$$\lim_{\delta x\to 0}\sum_{x=-8}^{-4}(\tfrac{4}{x} - 1)\delta x = \int_{-8}^{-4}(\tfrac{4}{x} - 1)dx = [4\ln|x| - x]_{-8}^{-4}$$
$$= (4\ln|-4| - (-4)) - (4\ln|-8| - (-8))$$
$$= (4\ln 4 + 4) - (4\ln 8 + 8) = 4(\ln 4 - \ln 8) - 4$$
$$= 4\ln\frac{4}{8} - 4 = 4\ln\frac{1}{2} - 4.$$

33 b)

$$\lim_{\delta x\to 0}\sum_{x=e}^{e^2}(\tfrac{3}{2x} + 12x)\delta x = \int_{e}^{e^2}(\tfrac{3}{2}\times\tfrac{1}{x} + 12x)dx = [\tfrac{3}{2}\ln|x| + 6x^2]_{e}^{e^2}$$
$$= (\tfrac{3}{2}\ln|e^2| + 6(e^2)^2) - (\tfrac{3}{2}\ln|e| + 6(e)^2)$$
$$= (\tfrac{3}{2}\ln e^2 + 6e^4) - (\tfrac{3}{2}\ln e + 6e^2)$$
$$= (\tfrac{3}{2}\times 2 + 6e^4) - (\tfrac{3}{2}\times 1 + 6e^2) = \frac{3}{2} + 6e^4 - 6e^2.$$

34)

$$\int_{-3}^{-1}\frac{10}{x}dx = [10\ln|x|]_{-3}^{-1} = 10\ln|-1| - 10\ln|-3|$$
$$= 10\ln 1 - 10\ln 3 = 0 - 10\ln 3 = -10\ln 3 = 10\ln 3^{-1} = 10\ln\frac{1}{3}.$$

$\therefore$ Area $= -10\ln\frac{1}{3} = 10\ln 3.$

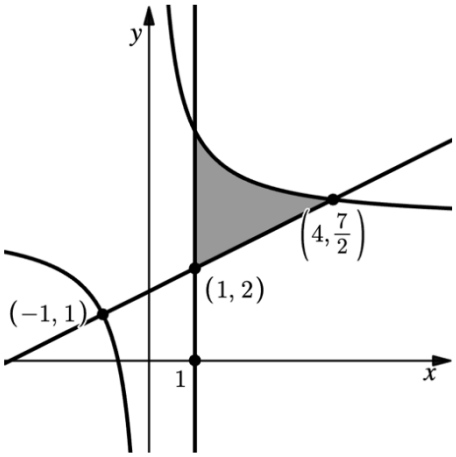
35)

$\dfrac{9-3x}{x} = 0 \Rightarrow 9-3x = 0 \Rightarrow x = 3.$ $\dfrac{3-3x}{x} = 0 \Rightarrow 3-3x = 0 \Rightarrow x = 1.$

$$\begin{aligned}\int_1^3 \frac{9-3x}{x}dx - \int_1^3 \frac{3-3x}{x}dx &= \int_1^3 \left(\frac{9-3x}{x} - \frac{3-3x}{x}\right)dx \\ &= \int_1^3 (\tfrac{9}{x} - \tfrac{3x}{x} - \tfrac{3}{x} + \tfrac{3x}{x})dx = \int_1^3 (\tfrac{9}{x} - 3 - \tfrac{3}{x} + 3)dx \\ &= \int_1^3 \frac{6}{x}dx = [6\ln|x|]_1^3 = 6\ln|3| - 6\ln|1| \\ &= 6\ln 3 - 6\ln 1 = 6\ln 3 - 6\times 0 = 6\ln 3.\end{aligned}$$

36)

$y = 3 + \frac{2}{x} \Rightarrow y = 3 + 2x^{-1} \Rightarrow \dfrac{dy}{dx} = -2x^{-2}.$ $\left.\dfrac{dy}{dx}\right|_{x=-1} = -2\times(-1)^{-2} = -2.$ Normal: $y - 1 = \frac{1}{2}(x + 1) \Rightarrow y = \frac{1}{2}x + \frac{3}{2}.$ $3 + \frac{2}{x} = \frac{1}{2}x + \frac{3}{2} \Rightarrow 6x + 4 = x^2 + 3x \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0 \Rightarrow x = -1$ or $x = 4.$ When $x = 4$, $y = \frac{1}{2}\times 4 + \frac{3}{2} = \frac{7}{2}.$ When $x = 1$, $y = \frac{1}{2}\times 1 + \frac{3}{2} = 2.$



$$\begin{aligned}\text{Area} &= \int_1^4 (3 + \tfrac{2}{x})dx - \frac{1}{2}\times 3\times(2 + \tfrac{7}{2}) \\ &= [3x + 2\ln|x|]_1^4 - \frac{33}{4} \\ &= (3\times 4 + 2\ln 4) - (3\times 1 + 2\ln 1) - \frac{33}{4} \\ &= (12 + 4\ln 2) - 3 - \frac{33}{4} = \frac{3}{4} + 4\ln 2.\end{aligned}$$

37 a)

$\int \sec^2 x\,dx = \tan x + c.$

37 b)

$\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + c.$

37 c)

$\int \sec x \tan x \, dx = \sec x + c.$

37 d)

$\int \operatorname{cosec}^2 x \, dx = -\cot x + c.$

37 e)

$\int \sec^2 4x \, dx = \frac{1}{4} \tan 4x + c.$

37 f)

$\int \operatorname{cosec}^2 \frac{1}{3}x \, dx = -3 \cot \frac{1}{3}x + c.$

38 a)

$\frac{dy}{dx} = \operatorname{cosec} 2x \cot 2x \Rightarrow y = -\frac{1}{2} \operatorname{cosec} 2x + c.$

38 b)

$\frac{dy}{dx} = 8 \operatorname{cosec} \frac{1}{4}x \cot \frac{1}{4}x \Rightarrow y = 8 \times -4 \operatorname{cosec} \frac{1}{4}x + c \Rightarrow y = -32 \operatorname{cosec} \frac{1}{4}x + c.$

38 c)

$\frac{dy}{dx} = \frac{2}{3} \sec \frac{10}{3}x \tan \frac{10}{3}x \Rightarrow y = \frac{2}{3} \times \frac{3}{10} \sec \frac{10}{3}x + c \Rightarrow y = \frac{1}{5} \sec \frac{10}{3}x + c.$

39)

$\frac{dy}{dx} = 5 \sec^2 3x \Rightarrow y = 5 \times \frac{1}{3} \tan 3x + c \Rightarrow y = \frac{5}{3} \tan 3x + c$. When $x = \frac{\pi}{12}$, $y = 5$: $5 = \frac{5}{3} \tan(3 \times \frac{\pi}{12}) + c \Rightarrow 5 = \frac{5}{3} \times 1 + c \Rightarrow c = \frac{10}{3}$. $\therefore y = \frac{5}{3} \tan 3x + \frac{10}{3}$.

40)

$\frac{dy}{dx} = \sec \frac{1}{2}x \tan \frac{1}{2}x \Rightarrow y = 2 \sec \frac{1}{2}x + c$. When $x = \frac{\pi}{3}$, $y = 0$: $0 = 2 \sec(\frac{1}{2} \times \frac{\pi}{3}) + c \Rightarrow 0 = \dfrac{2}{\cos \frac{\pi}{6}} + c \Rightarrow 0 = \frac{4\sqrt{3}}{3} + c \Rightarrow c = -\frac{4\sqrt{3}}{3}$. $\therefore y = 2 \sec \frac{1}{2}x - \frac{4\sqrt{3}}{3}$.

41 a)

$$\int_0^{\pi/6} 4 \sec^2 2x \, dx = [4 \times \tfrac{1}{2} \tan 2x]_0^{\pi/6} = [2 \tan 2x]_0^{\pi/6}$$
$$= 2 \tan(2 \times \tfrac{\pi}{6}) - 2 \tan(2 \times 0) = 2 \tan \frac{\pi}{3} - 2 \tan 0 = 2\sqrt{3} - 0 = 2\sqrt{3}.$$

41 b)

$$\int_{\pi/2}^{3\pi/2} 9 \operatorname{cosec} \tfrac{1}{3}x \cot \tfrac{1}{3}x \, dx = [9 \times -3 \operatorname{cosec} \tfrac{1}{3}x]_{\pi/2}^{3\pi/2} = [-27 \operatorname{cosec} \tfrac{1}{3}x]_{\pi/2}^{3\pi/2}$$
$$= -27 \operatorname{cosec}(\tfrac{1}{3} \times \tfrac{3\pi}{2}) - (-27 \operatorname{cosec}(\tfrac{1}{3} \times \tfrac{\pi}{2}))$$
$$= -\frac{27}{\sin \frac{\pi}{2}} - (-\frac{27}{\sin \frac{\pi}{6}}) = -27 - (-54) = 27.$$

41 c)

$$\begin{aligned}\int_{\pi/6}^{\pi/4} \tfrac{1}{3}\operatorname{cosec}^2 3x\,dx &= [\tfrac{1}{3}\times -\tfrac{1}{3}\cot 3x]_{\pi/6}^{\pi/4} = [-\tfrac{1}{9}\cot 3x]_{\pi/6}^{\pi/4}\\ &= -\frac{1}{9}\cot(3\times\tfrac{\pi}{4}) - (-\frac{1}{9}\cot(3\times\tfrac{\pi}{6}))\\ &= -\frac{1}{9}\times\frac{\cos\frac{3\pi}{4}}{\sin\frac{3\pi}{4}} - (-\frac{1}{9}\times\frac{\cos\frac{\pi}{2}}{\sin\frac{\pi}{2}})\\ &= -\frac{1}{9}\times -1 - (-\frac{1}{9}\times 0) = \frac{1}{9} - 0 = \frac{1}{9}.\end{aligned}$$

41 d)

$$\begin{aligned}\int_{-\pi/4}^{0} -\sec\tfrac{2}{3}x\tan\tfrac{2}{3}x\,dx &= [-\tfrac{3}{2}\sec\tfrac{2}{3}x]_{-\pi/4}^{0}\\ &= -\frac{3}{2}\sec(\tfrac{2}{3}\times 0) - (-\frac{3}{2}\sec(\tfrac{2}{3}\times(-\tfrac{\pi}{4})))\\ &= -\frac{3}{2}\times\frac{1}{\cos 0} - (-\frac{3}{2}\times\frac{1}{\cos(-\frac{\pi}{6})})\\ &= -\frac{3}{2} - (-\sqrt{3}) = \sqrt{3} - \frac{3}{2}.\end{aligned}$$

42 a)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=-\pi/2}^{\pi/2}\tfrac{5}{4}\sec^2\tfrac{5}{6}x\,\delta x &= \int_{-\pi/2}^{\pi/2}\tfrac{5}{4}\sec^2\tfrac{5}{6}x\,dx = [\tfrac{5}{4}\times\tfrac{6}{5}\tan\tfrac{5}{6}x]_{-\pi/2}^{\pi/2}\\ &= [\tfrac{3}{2}\tan\tfrac{5}{6}x]_{-\pi/2}^{\pi/2}\\ &= \frac{3}{2}\tan(\tfrac{5}{6}\times\tfrac{\pi}{2}) - \frac{3}{2}\tan(\tfrac{5}{6}\times(-\tfrac{\pi}{2}))\\ &= \frac{3}{2}(2+\sqrt{3}) - \frac{3}{2}(-2-\sqrt{3})\\ &= \frac{3}{2}((2+\sqrt{3}) - (-2-\sqrt{3})) = \frac{3}{2}(4+2\sqrt{3}) = 3(2+\sqrt{3}).\end{aligned}$$

42 b)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=\pi/9}^{2\pi/9}\tfrac{3}{8}\sec\tfrac{3}{2}x\tan\tfrac{3}{2}x\,\delta x &= \int_{\pi/9}^{2\pi/9}\tfrac{3}{8}\sec\tfrac{3}{2}x\tan\tfrac{3}{2}x\,dx = [\tfrac{3}{8}\times\tfrac{2}{3}\sec\tfrac{3}{2}x]_{\pi/9}^{2\pi/9}\\ &= [\tfrac{1}{4}\sec\tfrac{3}{2}x]_{\pi/9}^{2\pi/9}\\ &= \frac{1}{4}\sec(\tfrac{3}{2}\times\tfrac{2\pi}{9}) - \frac{1}{4}\sec(\tfrac{3}{2}\times\tfrac{\pi}{9})\\ &= \frac{1}{4}\times\frac{1}{\cos\frac{\pi}{3}} - \frac{1}{4}\times\frac{1}{\cos\frac{\pi}{6}} = \frac{1}{2} - \frac{\sqrt{3}}{6}.\end{aligned}$$

43)

$$\begin{aligned}\int_{-\pi/16}^{\pi/16}(\sec^2 4x-2)dx &= [\tfrac{1}{4}\tan 4x-2x]_{-\pi/16}^{\pi/16}\\ &= (\tfrac{1}{4}\tan(4\times\tfrac{\pi}{16})-2\times\tfrac{\pi}{16})\\ &\quad -(\tfrac{1}{4}\tan(4\times(-\tfrac{\pi}{16}))-2\times(-\tfrac{\pi}{16}))\\ &= (\tfrac{1}{4}-\tfrac{\pi}{8})-(-\tfrac{1}{4}+\tfrac{\pi}{8}) = \frac{1}{2}-\frac{\pi}{4}.\end{aligned}$$

$\therefore$ Area $= \frac{\pi}{4}-\frac{1}{2}$.

44)

$$\begin{gathered}\operatorname{cosec}x\cot x = \sec x\tan x,\\ \frac{1}{\sin x}\times\frac{\cos x}{\sin x} = \frac{1}{\cos x}\times\frac{\sin x}{\cos x}\Rightarrow\frac{\cos x}{\sin^2 x} = \frac{\sin x}{\cos^2 x},\\ \cos^3 x = \sin^3 x\Rightarrow\frac{\sin^3 x}{\cos^3 x} = 1\Rightarrow\tan^3 x = 1,\\ \tan x = 1\Rightarrow x = \arctan 1 = \frac{\pi}{4}.\end{gathered}$$

$$\begin{aligned}\int_{\pi/6}^{\pi/4}\operatorname{cosec}x\cot x\,dx-\int_{\pi/6}^{\pi/4}\sec x\tan x\,dx &= \int_{\pi/6}^{\pi/4}(\operatorname{cosec}x\cot x-\sec x\tan x)dx\\ &= [-\operatorname{cosec}x-\sec x]_{\pi/6}^{\pi/4}\\ &= (-\operatorname{cosec}\tfrac{\pi}{4}-\sec\tfrac{\pi}{4})-(-\operatorname{cosec}\tfrac{\pi}{6}-\sec\tfrac{\pi}{6})\\ &= (-\tfrac{1}{\sin\frac{\pi}{4}}-\tfrac{1}{\cos\frac{\pi}{4}})-(-\tfrac{1}{\sin\frac{\pi}{6}}-\tfrac{1}{\cos\frac{\pi}{6}})\\ &= (-\sqrt{2}-\sqrt{2})-(-2-\tfrac{2\sqrt{3}}{3}) = -2\sqrt{2}+2+\frac{2\sqrt{3}}{3}.\end{aligned}$$

45)

$y = \operatorname{cosec}^2 x \Rightarrow y = (\sin x)^{-2} \Rightarrow \dfrac{dy}{dx} = \cos x\times -2(\sin x)^{-3} \Rightarrow \dfrac{dy}{dx} = -\dfrac{2\cos x}{\sin^3 x}.$ $\left.\dfrac{dy}{dx}\right|_{x=\pi/4} = -\dfrac{2\cos\frac{\pi}{4}}{\sin^3\frac{\pi}{4}} = -4.$ When $x=\frac{\pi}{4}$, $y = \operatorname{cosec}^2\frac{\pi}{4} = \dfrac{1}{\sin^2\frac{\pi}{4}} = 2$. So $B(\frac{\pi}{4},2)$. Tangent at B is $y-2 = -4(x-\frac{\pi}{4}) \Rightarrow y = -4x+\pi+2$. When $x=0$, $y=\pi+2$. So $A(0,\pi+2)$. Area of trapezium $= \frac{1}{2}\times\frac{\pi}{4}\times((\pi+2)+2) = \frac{\pi}{8}(\pi+4) = \frac{\pi^2+4\pi}{8}$.

$$\begin{aligned}\int_{\pi/4}^{\pi/2}\operatorname{cosec}^2 x\,dx &= [-\cot x]_{\pi/4}^{\pi/2} = (-\tfrac{\cos\frac{\pi}{2}}{\sin\frac{\pi}{2}})-(-\tfrac{\cos\frac{\pi}{4}}{\sin\frac{\pi}{4}})\\ &= 0-(-1) = 1.\end{aligned}$$

$\therefore$ Area $= \dfrac{\pi^2+4\pi}{8}+1 = \dfrac{\pi^2+4\pi+8}{8}$.